Charles Benello

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Education

The University of Chicago — Chicago, IL
Master of Science in Financial Mathematics — *Expected Mar 2027*
- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: Distributed Systems, ML in Software Engineering, Agentic AI, Reinforcement Learning

The University of Chicago | GPA: 3.75/4.0 — Chicago, IL
Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean — *Jun 2025*
- Coursework: Database Systems, Operating Systems, Algorithms, Concurrency Control, Computer Architecture, Networks

Experience

Incoming Software Development Engineer Intern — Aug – Nov 2026
Amazon Web Services (Redshift) — *Palo Alto, CA*
- Joining the Amazon Redshift query optimization team to work on plan-time decisions and cost models for analytical workloads.

Database Research Assistant — Apr 2023 – Present
ChiData Research Lab, University of Chicago — *Chicago, IL*
- First-authored VLDB 2026 paper (CrocSort) by designing a parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort, jointly planning memory and phase-specific thread counts.
- Co-authored CIDR 2025 paper on resource-adaptive query execution with paged memory management.
- Sole-authored HPTS 2026 paper on a lightweight prediction model for sort configuration on cloud and disaggregated hardware, transferring calibration across machines within 3–5% of optimal without brute-force sweeps.

GPU Systems Researcher — Jun – Sep 2025
EPFL (DIAS Lab, Prof. Anastasia Ailamaki) — *Lausanne, Switzerland*
- Reduced out-of-memory query times by 50–200% on large TPC-DS analytical workloads by engineering a macroblock-based GPU decompression strategy in CUDA/C++ that pipelines compressed column blocks through HBM, eliminating full-column staging and unlocking queries on datasets that exceed device memory capacity.
- Enabled adaptive decompression strategy selection at plan time by building cost-based simulation models predicting query performance across workload complexities, validated on TPC-DS at multiple scale factors with mixed compression schemes.
- Identified GPU pipeline bottlenecks under concurrent workloads by designing TPC-DS join patterns and resource contention models across I/O, decompression, and compute stages.
- Shaped a new cost-based research direction (with a forthcoming publication) by benchmarking GPU compression trade-offs across schemes and dataset characteristics.

Software Engineering Intern — Feb 2026 – Present
Alter Domus — *Chicago, IL*
- Lifted document extraction accuracy from 50% to 92–97% via a multi-stage pipeline for 100+ page financial filings.
- Built an LLM-as-a-Judge evaluation framework benchmarking faithfulness, numerical accuracy, and completeness across document types.

Quantitative Researcher — Mar 2026 – Present
PayPal (UChicago Project Lab) — *Chicago, IL*
- Pretrained a TabBERT transaction foundation model on 74M MBD transactions (Sberbank, 200K users) and fine-tuned cross-dataset to IBM TabFormer; partial fine-tuning lifted FM AUC from 0.74 to 0.996, with stacked XGBoost+FM saving $215K per 2K-user test cohort.
- Audited the original 16-feature XGBoost baseline (AUC 0.66) and built a 21-feature replacement reaching AUC 0.99; built 14 fusion variants (stacking, isotonic calibration, MLP) and per-user diagnostic dashboards (t-SNE, kNN, KMeans).

Projects

GPU External Sort | *CUDA, C++* | github.com/cmbenello/gpu-research — 2025–Present
- Built a GPU external merge sort in CUDA (port of CrocSort, VLDB 2026). On TPC-H SF50 ORDER BY lineitem (300M rows, 36 GB), beats DuckDB by 20.1×, Polars by 34.5×, and MendSort (JouleSort 2023) by 2.4× on a PCIe-Gen3 GPU. Cost-based plan-time selector between compact-key and full-key paths cuts PCIe amplification to 0.14× via key-only upload, permutation gather, and triple-buffered streaming.

Transit Coverage Optimizer | *Python* | cmbenello.github.io/covtsp — 2026
- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) with an automated Covering TSP pipeline on a time-expanded graph (160K nodes, 1.26M edges) using k-NN with urgency scoring and epsilon-greedy search. Generalized to NYC (475 stations) and Berlin.

Concurrent Database Engine | *Rust* | github.com/cmbenello/CrustyDB — 2025
- Built a multi-threaded relational database in Rust: B+ tree indexing, concurrent buffer pool, heap storage, SQL execution.

Technical Skills

Languages: C/C++, Python, Rust, CUDA, Go, SQL, TypeScript, JavaScript, Bash
Tools: PostgreSQL, DuckDB, Databricks, Docker, Git, Linux, PyTorch, FastAPI, Next.js, SLURM
Concepts: Distributed Systems, Database Internals, Query Execution & Optimization, Concurrency, GPU Programming, Performance Engineering, Cost-Based Optimization, System Design

Publications

[1] R. Otaki*, **C. Benello***, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026.* (*Equal contribution)
[2] **C. Benello**. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026.*
[3] R. Otaki, J.H. Chang, **C. Benello**, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution with Paged Memory Management." *CIDR 2025.*

Additional

Languages: Korean, Italian (Intermediate); French (Basic) | **Interests**: Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting